AAE-E2001 Computational Fluid Dynamics D
Assignment 2: Lid-driven cavity

Theory exercise 20/100

For the linear 1D convection diffusion equation

$$\frac{\partial c}{\partial t} + u\frac{\partial c}{\partial x} = \Gamma\frac{\partial^2 c}{\partial x^2} \tag{1}$$

show that the semi-discrete form obtained with the finite-volume method corresponds to the one obtained with the finite-difference method

$$\frac{\partial c_i}{\partial t} = -u\left(\frac{c_{i+1} - c_{i-1}}{2\Delta x}\right) + \Gamma\left(\frac{c_{i+1} - 2c_i + c_{i-1}}{\Delta x^2}\right) \tag{2}$$

if linear interpolation is used.

Hint: start from the conservative form

$$\frac{\partial c}{\partial t} = -\frac{\partial f_C}{\partial x} + \frac{\partial f_D}{\partial x}, \tag{3}$$

where $f_C = uc$ is the convection flux and $f_D = \Gamma\frac{\partial c}{\partial x}$ is the diffusion flux and apply the Gauss theorem. Assume the following:

- Constant $\Delta x, u$ and Γ
- Integrals are computed with the midpoint rule $\int_{S_{i+1/2}} f\, dS \approx f_{i+1/2}S_{i+1/2}$
- Derivative in the diffusion flux is computed with the second-order central difference

Simulation exercise 80/100

The lid-driven cavity flow case is a simple set-up where the fluid is contained in a square-shaped cavity and is driven by the motion of the upper wall, the lid. The Reynolds number is defined using the velocity of the lid U, the cavity edge length L, and the kinematic viscosity of the fluid ν, i.e.

$$Re = \frac{UL}{\nu}.$$

The set-up is shown in Figure 1 together with the relevant parameters. After an initial transient period, the flow will reach a steady state if the Reynolds number is small enough to maintain laminar condition.

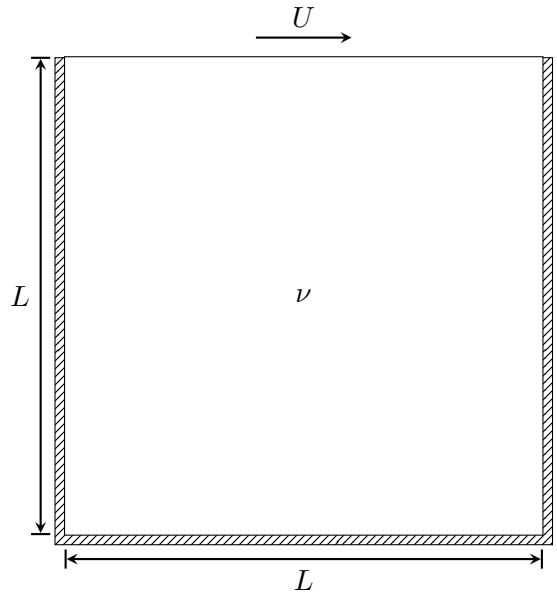


Figure 1: Lid-driven cavity flow.

Prepare three lid-driven flow cases at $Re = 100$ using the following mesh resolutions: $n_x \times n_y = 10 \times 10, 40 \times 40$, and 80×80. In your report, mention which values you chose for the parameters shown in Figure 1 together with the boundary conditions and the initial conditions. Use `incompressibleFluid` solver. A good starting point for your cavity set-up can be found in the OpenFOAM tutorial:
`$FOAM_TUTORIALS/incompressibleFluid/cavity`.

Copy that to your working directory and make changes as needed. Run the flow cases you set up using the `incompressibleFluid` solver with `nOuterCorrectors` set to 1 and `nCorrectors` set to two or higher (Note that your are using PIMPLE but with these settings, it is running in the PISO mode). Simulations should be run for a time period of $T = 10L/U$ and with the same Courant number for all cases.

In your report, present the following figures:

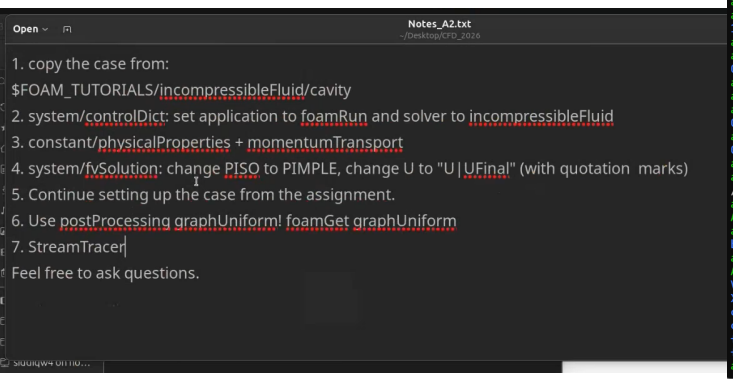
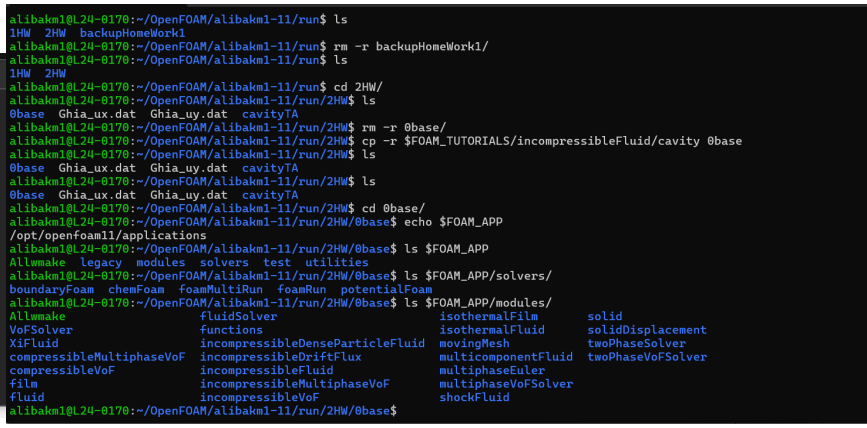
- Mesh picture for one of the cases.
- Streamlines drawn at the latest time instance from all three simulations (Three figures).
- Horizontal velocity of the 40×40 simulation on a vertical line going through the center of the geometry i.e. u_x vs y from 5 different time instances compared to the results of Ghia et al. (Single figure).
- Vertical velocity of the 40×40 simulation on a horizontal line going

through the center of the geometry i.e. u_y vs x from 5 different time instances compared to the results of Ghia et al. (Single figure).

- Horizontal velocity at the latest time instance of all three simulations on a vertical line going through the center of the geometry i.e. u_x vs y compared to the results of Ghia et al. (Single figure).
- Vertical velocity at the latest time instance of all three simulations on a horizontal line going through the center of the geometry i.e. u_y vs x compared to the results of Ghia et al. (Single figure).

Based on your results, answer to the following questions:

1. How would you describe the flow field obtained what kinds of features can be distinguished?
2. How does the flow develop in time?
3. How does increasing the mesh resolution affect the results? Comment on the grid-independent nature of the results?

References

[1] U. Ghia, K. N. Ghia, and C. T. Shin: "High-Re solutions for incompressible flow using the Navier-Stokes equations and a multigrid method", Journal of Computational Physics **3**, 387–411, (1982).